



Java SE 8 Programming Language
Practice Final Test

Document Code	25e-BM/HR/HDCV/FSOFT
Version	1.0
Effective Date	20/11/2012

Hanoi, 08/2023

RECORD OF CHANGES

No	Effective Date	Change Description	Reason	Reviewer	Approver



CODE:	FR-IVI_JAVA_Practice_T01
TYPE:	LONG
LOC:	N/A
DURATION:	180 Minutes

Exam 1:

Write a program to input a positive integer n and perform the following functions:

- Calculate the sum of the digits of n .
- Factorize n into prime factors.
- List the divisors of n .
- List the prime divisors of n .

Exam 2:

Write a program to list the integers with 5 to 7 digits that satisfy:

- Being a prime number.
- Being a palindrome number. (ex 77377, 3223223...)
- Having all digits as prime numbers.
- Having a prime number as the sum of its digits. (ex The sum of 5-digit prime numbers is: 8363)

Exam 3:

Input an array of integers $a_0, a_1, a_2, \dots, a_{n-1}$. Sort the array in ascending order, then insert element x while ensuring the array remains in increasing order.

Example

input = [7, 5, 3, 1, 4, 8]

The element to be inserted is 6

after doing the sort

output = [1, 3, 4, 5, 6, 7, 8]

- Data will be entered by the user

Exam 4:

Build an object named "**Contact Management**" as follows:

Build a "**Contact**" class:

Attributes: Name, Phone Number

Methods:

Constructor method to input data for the "Name" and "Phone Number" attributes of a contact.

Get/Set methods for attributes.

Build a "**Management**" class:

Attributes: List of contacts

Methods:

- Check if a given phone number exists in the contact list. If it does, return true; otherwise, return false.
- Add a contact (name, phone number):
 - Before adding a contact, check if the phone number already exists. If it does, display the message "Contact already exists." If not, add the contact to the list and display "Contact added successfully."
- Edit a contact (old phone number, new phone number):
 - To edit the phone number in the contact list, two conditions need to be checked:
 - The old phone number must exist in the contact list.
 - The new phone number must not already exist in the contact list.
 - If both conditions are met, update the contact's phone number and display a message.
- Search for a contact (name):
 - Search for all contacts with names matching the input and display their information.
 - For example, if "Hoang" is entered, display contacts like "Hoang," "VuHoang," "ToHoang," etc.
- Sort contacts alphabetically.

Requirements:

Use ArrayList knowledge to construct object methods.

-- THE END --